Light

GCSE Computer Science - Memory and Storage (Set 2)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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← Back to Memory And Storage

Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

Which statement correctly describes a key characteristic of RAM?

- **A.** RAM is volatile and loses its contents when power is switched off
- **B.** RAM retains data permanently even when power is removed
- **C.** RAM can only be read from, never written to
- **D.** RAM stores the computer's startup instructions

Q2 (1 mark)

What happens to the data stored in ROM when a computer is switched off?

- **A.** The data is automatically backed up to the cloud

- ○ **B.** The data is transferred to RAM for safekeeping
- ○ **C.** The data is permanently deleted
- ○ **D.** The data remains intact because ROM is non-volatile

Q3 (1 mark)

What does 'non-volatile' mean when describing computer storage?

- ○ **A.** The storage device operates silently without noise
- ○ **B.** The storage cannot be removed from the computer
- ○ **C.** The storage retains data without requiring continuous power
- ○ **D.** The storage is resistant to computer viruses

Q4 (1 mark)

Which of the following is an example of volatile memory?

- ○ **A.** RAM
- ○ **B.** Hard disk drive
- ○ **C.** Solid state drive
- ○ **D.** USB flash drive

Q5 (1 mark)

What is the primary reason computers need secondary storage?

- ○ **A.** To make calculations faster
- ○ **B.** To display images on the screen
- ○ **C.** To increase the processor speed
- ○ **D.** To store data permanently when the power is off

Q6 (1 mark)

Where is cache memory typically located in a computer system?

- ○ **A.** On or very close to the CPU
- ○ **B.** On the hard disk drive
- ○ **C.** Inside the power supply unit
- ○ **D.** On the network card

Q7 (1 mark)

Why are registers the fastest type of memory in a computer?

- ○ **A.** Because they are built directly into the CPU with minimal access distance
- ○ **B.** Because they have the largest storage capacity
- ○ **C.** Because they use magnetic storage technology
- ○ **D.** Because they are connected to the internet

Q8 (1 mark)

Which components are found inside a magnetic hard disk drive?

- ○ **A.** Spinning platters and read/write heads
- ○ **B.** Flash memory chips and controllers
- ○ **C.** Optical lasers and reflective discs
- ○ **D.** Radio transmitters and receivers

Q9 (1 mark)

What is a key physical characteristic that distinguishes SSDs from HDDs?

- ○ **A.** SSDs are always larger in physical size
- ○ **B.** SSDs can only store text files
- ○ **C.** SSDs must be connected to the internet to function
- ○ **D.** SSDs contain no moving mechanical parts

Q10 (1 mark)

What is required to access files stored in cloud storage?

- ○ **A.** A DVD drive
- ○ **B.** A magnetic tape reader
- ○ **C.** A floppy disk drive

- ○ **D.** An internet connection

Q11 (1 mark)

Which statement about RAM operations is correct?

- ○ **A.** RAM can only be written to once
- ○ **B.** RAM can only be read from, never written to
- ○ **C.** RAM can be both read from and written to many times
- ○ **D.** RAM deletes data automatically every hour

Q12 (1 mark)

Why is ROM described as 'read only' during normal computer operation?

- ○ **A.** Because ROM cannot store any data at all
- ○ **B.** Because ROM operates only when the computer is off
- ○ **C.** Because ROM is invisible to the operating system
- ○ **D.** Because data in ROM can be read but not modified during normal use

Q13 (2 marks)

A user notices their computer becomes very slow when they have many browser tabs and applications open. How would increasing the amount of RAM help?

- ○ **A.** It would make the internet connection faster
- ○ **B.** It would make the monitor display images more quickly
- ○ **C.** It would increase the size of the hard drive
- ○ **D.** It would allow more programs and data to be held in fast memory, reducing the need for slow virtual memory access

Q14 (2 marks)

Why do computers with SSDs typically start up faster than those with HDDs?

- ○ **A.** SSDs have larger storage capacity
- ○ **B.** SSDs connect to faster internet services
- ○ **C.** SSDs automatically compress the operating system
- ○ **D.** SSDs access data electronically without waiting for mechanical parts to move, enabling faster read speeds

Q15 (2 marks)

A film studio needs to store 100TB of raw video footage for archival purposes. Why might they choose HDDs over SSDs?

- ○ **A.** HDDs provide significantly lower cost per terabyte, making large-scale storage affordable
- ○ **B.** HDDs are faster at reading video files
- ○ **C.** HDDs are more portable for carrying between locations

- ○ **D.** HDDs produce better video quality when played back

Q16 (2 marks)

A team of designers working in different countries needs to collaborate on the same project files. Why is cloud storage particularly suitable for this situation?

- ○ **A.** Cloud storage makes files download faster than local storage
- ○ **B.** Cloud storage allows all team members to access the same files from any location with internet, enabling real-time collaboration
- ○ **C.** Cloud storage automatically improves the quality of design files
- ○ **D.** Cloud storage eliminates the need for computers entirely

Q17 (2 marks)

A business relies entirely on cloud storage for all their files. What significant risk does this create?

- ○ **A.** The files will automatically become larger over time
- ○ **B.** If their internet connection fails or the cloud service experiences an outage, they cannot access any of their data
- ○ **C.** Cloud storage causes computers to overheat
- ○ **D.** The business will need to hire more staff

Q18 (2 marks)

A computer has 8MB of cache memory. During operation, the CPU needs data that is not currently in the cache. What happens?

- ○ **A.** The CPU stops working until the computer is restarted
- ○ **B.** The data is downloaded from the internet
- ○ **C.** The CPU retrieves the data from slower main memory (RAM) and a copy may be placed in cache for faster future access
- ○ **D.** The cache automatically expands to store more data

Q19 (2 marks)

Which statement correctly compares the characteristics of registers, RAM, and secondary storage?

- ○ **A.** Registers are fastest with smallest capacity, RAM is medium speed with medium capacity, secondary storage is slowest with largest capacity
- ○ **B.** Registers have the largest capacity, RAM has medium capacity, secondary storage has smallest capacity
- ○ **C.** Secondary storage is fastest, RAM is medium speed, registers are slowest
- ○ **D.** All three have identical speed and capacity characteristics

Q20 (2 marks)

A delivery driver uses a laptop in their vehicle, which experiences frequent bumps and vibrations. Why would an SSD be more suitable than an HDD for this laptop?

- ○ **A.** SSDs are waterproof while HDDs are not
- ○ **B.** SSDs generate more heat to keep the laptop warm

- ○ **C.** SSDs have no moving parts, so they are resistant to damage from shock and vibration
- ○ **D.** SSDs connect to satellite navigation systems

Q21 (2 marks)

When a user opens a word processor application, what role does main memory (RAM) play?

- ○ **A.** RAM permanently stores the word processor for future use
- ○ **B.** RAM holds the word processor code and current document while the application is running, allowing fast CPU access
- ○ **C.** RAM displays the word processor on screen
- ○ **D.** RAM saves the document to the hard drive

Q22 (2 marks)

Why must the initial startup instructions be stored in ROM rather than RAM?

- ○ **A.** ROM is faster than RAM
- ○ **B.** RAM is empty when power is first applied because it is volatile, so non-volatile ROM must provide the first instructions
- ○ **C.** ROM has more storage capacity than RAM
- ○ **D.** RAM cannot be read by the processor

Q23 (2 marks)

What is virtual memory and when is it used?

- ○ **A.** Memory that only exists in video games
- ○ **B.** A section of the hard drive used as overflow when RAM is full, allowing more programs to run than physical RAM could hold alone
- ○ **C.** A type of RAM that costs less than regular RAM
- ○ **D.** Memory that automatically backs up to the cloud

Q24 (3 marks)

A gaming enthusiast is building a new PC and wants the fastest possible game loading times, but also needs to store their large library of 200+ games (approximately 4TB total). With a limited budget, what storage configuration would best balance their needs?

- ○ **A.** A single 4TB HDD for all storage needs
- ○ **B.** A single 256GB SSD for all storage needs
- ○ **C.** A smaller SSD (500GB-1TB) for the operating system and frequently played games, combined with a larger HDD (4TB) for storing the full game library
- ○ **D.** Only cloud storage to avoid buying any drives

Q25 (3 marks)

A computer is running several applications when a sudden power failure occurs. Explain what data would be lost and what would be preserved, and why.

- ○ **A.** All data everywhere would be lost because computers cannot function without power

○ **B.** All data everywhere would be preserved because modern computers have backup systems

○ **C.** Unsaved work in RAM would be lost because RAM is volatile, but saved files on the SSD/HDD and BIOS in ROM would be preserved because they are non-volatile

○ **D.** Only files larger than 1GB would be lost due to size limitations

Q26 (3 marks)

A freelance video editor works from home and travels frequently to client sites. They need to access project files from multiple locations and share previews with clients, but work with very large video files (50-100GB per project). Analyse the trade-offs between cloud and local storage for their situation.

○ **A.** A hybrid approach: local SSD storage for active editing (cloud upload of 50GB+ files would be impractical for daily work), with cloud storage for sharing smaller preview files with clients and backing up completed projects

○ **B.** Local storage only, as cloud storage cannot handle video files

○ **C.** Cloud storage is perfect with no drawbacks for this use case

○ **D.** Neither cloud nor local storage would work for video editing

Q27 (3 marks)

Explain why increasing cache size improves performance up to a point, but has diminishing returns beyond a certain size.

○ **A.** Cache size has no effect on performance at any level

○ **B.** Larger cache always provides proportional performance increases with no limit

○ **C.** Cache performance depends only on the clock speed of the processor

○ **D.** Initially, larger cache captures more frequently-used data, reducing slow RAM accesses; beyond a certain point, the most frequently used data is already cached, and additional cache provides less benefit as it only captures rarely-used data

Q28 (3 marks)

Compare how data is physically stored in SSDs versus HDDs, and explain how these different mechanisms affect reliability and lifespan.

○ **A.** Both use identical magnetic storage; there is no difference in mechanism or reliability

○ **B.** SSDs store data as electrical charges in flash memory cells (no moving parts, resistant to shock but cells degrade with write cycles); HDDs store data magnetically on spinning platters (moving parts vulnerable to shock, but magnetic storage does not degrade with use)

○ **C.** SSDs use optical lasers while HDDs use radio waves; both are equally reliable

○ **D.** HDDs are electronic while SSDs are mechanical; HDDs last longer

Q29 (3 marks)

A computer has 4GB of RAM and is running applications that require 6GB total. Describe the memory management process the operating system uses and analyse its impact on performance.

○ **A.** The computer will refuse to run the applications because RAM is insufficient

○ **B.** The operating system uses virtual memory: it moves less-active data from RAM to the hard drive, freeing RAM for active needs; this

allows operation but causes slowdowns when swapped data is needed, as hard drive access is much slower than RAM

○ **C.** The applications automatically reduce their memory requirements to fit

○ **D.** The RAM physically expands to accommodate the extra requirement

Q30 (3 marks)

Trace the role of each memory and storage type during a computer's boot process, from power-on until the operating system is fully loaded.

○ **A.** RAM loads the operating system first, then transfers it to ROM for permanent storage

○ **B.** The hard drive directly executes the operating system without involving any memory

○ **C.** ROM provides initial boot instructions (BIOS/UEFI) that test hardware and locate the OS on secondary storage; the OS is then loaded from SSD/HDD into RAM where the CPU can execute it; cache and registers are used during this processing for speed

○ **D.** Cloud storage downloads the operating system fresh each time the computer starts

Total: **55** marks • Time: **30 min**

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